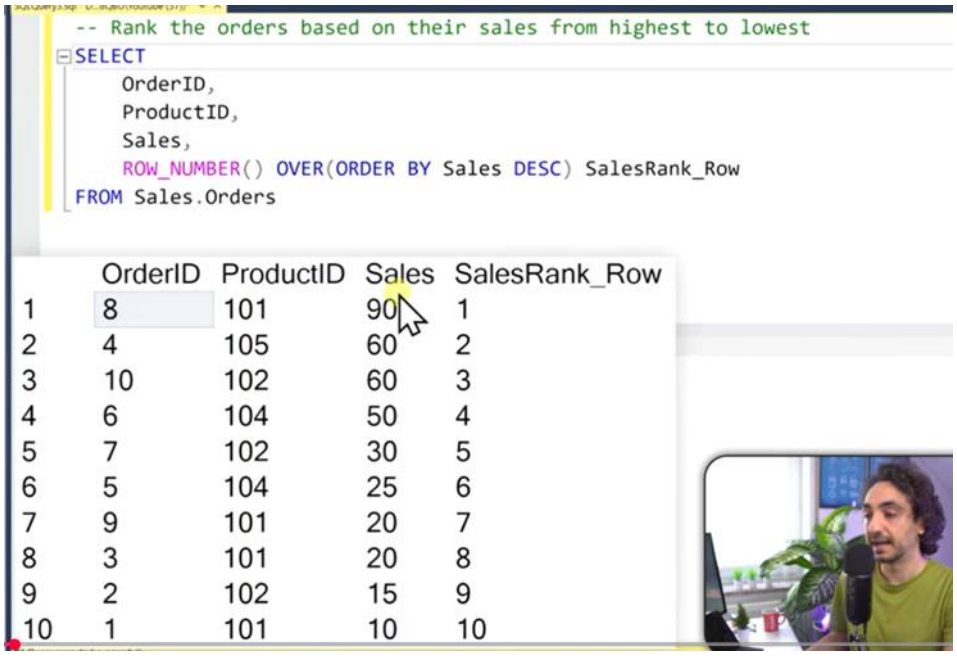
Window Function - Rank function

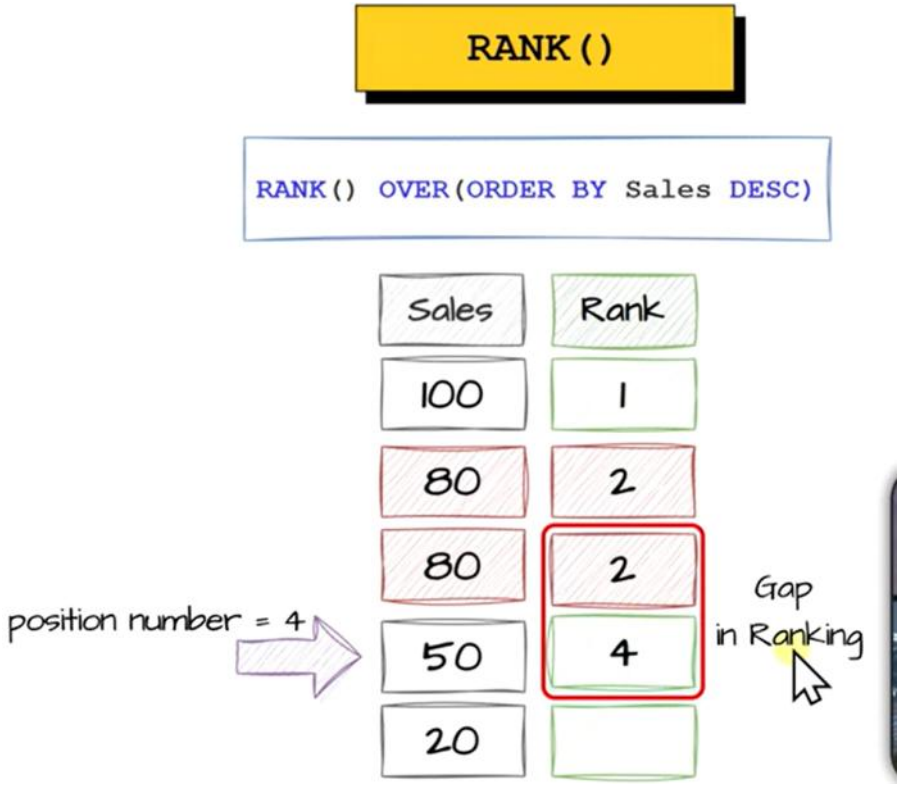
04 November 2026 10:45

ROW_NUMBER() - Assign a unique number to each row.
It doesn't handle ties.

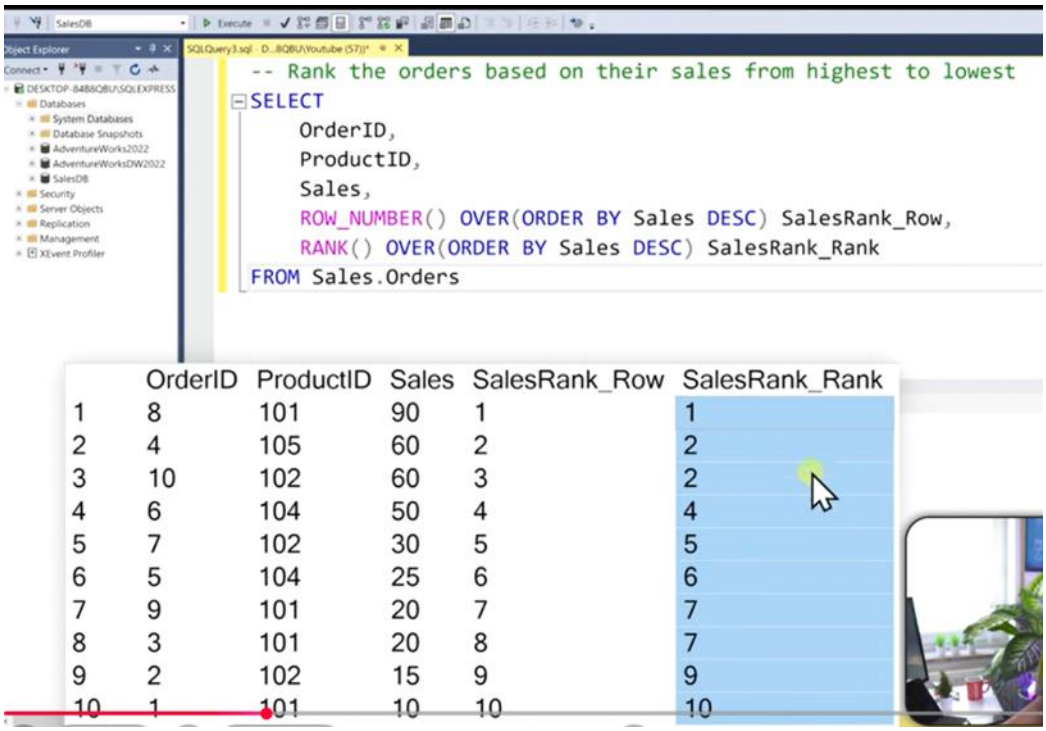




RANK():- Assign a rank to each row.
It handles ties.
It leaves gaps in ranking.

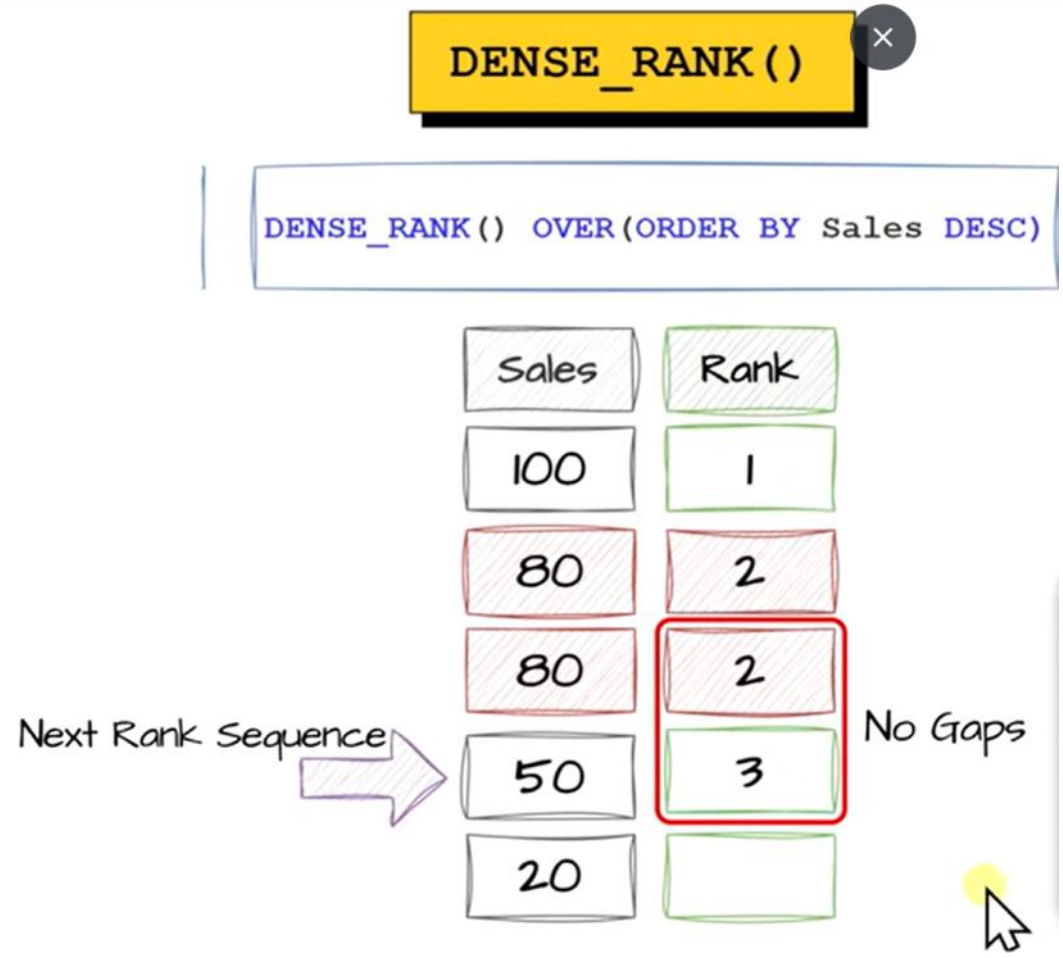


The next one will be rank - 5



DENSE_RANK() -

Assign a rank to each row.
It handles ties
It doesn't leaves gaps in ranking.




```sql
-- Rank the orders based on their sales from highest to lowest
SELECT
    OrderID,
    ProductID,
    Sales,
    ROW_NUMBER() OVER(ORDER BY Sales DESC) SalesRank_Row,
    RANK()       OVER(ORDER BY Sales DESC) SalesRank_Rank,
    DENSE_RANK() OVER(ORDER BY Sales DESC) SalesRank_Dense
FROM Sales.Orders
```

	OrderID	ProductID	Sales	SalesRank_Row	SalesRank_Rank	SalesRank_Dense
1	8	101	90	1	1	1
2	4	105	60	2	2	2
3	10	102	60	3	2	2
4	6	104	50	4	4	3
5	7	102	30	5	5	4
6	5	104	25	6	6	5
7	9	101	20	7	7	6
8	3	101	20	8	7	6
9	2	102	15	9	9	7
10	1	101	10	10	10	8

Below ranking is done for particular products:-

```sql
-- Find the top highest sales for each product

SELECT
    OrderID,
    ProductID,
    Sales,
    ROW_NUMBER() OVER (PARTITION BY ProductID ORDER BY Sales DESC) RankByProduct
FROM Sales.Orders
```

	OrderID	ProductID	Sales	RankByProduct
1	8	101	90	1
2	9	101	20	2
3	3	101	20	3
4	1	101	10	4
5	10	102	60	1
6	7	102	30	2
7	2	102	15	3
8	6	104	50	1
9	5	104	25	2
10	4	105	60	1



```sql
-- Find the lowest 2 customers based on their total sales
SELECT *
FROM (
SELECT
    CustomerID,
    SUM(Sales) TotalSales,
    ROW_NUMBER() OVER (ORDER BY SUM(Sales)) RankCustomers
FROM Sales.Orders
GROUP BY
CustomerID
)t WHERE RankCustomers <= 2
```

	CustomerID	TotalSales	RankCustomers
1	2	55	1
2	4	90	2

NTILE():-

Divides the rows in to a specified number of approximately equal groups(Buckets).

NTILE(2) OVER(ORDER BY SALES DESC).

Bucket Size = Number of rows/Number of Buckets(2 used here)

So if there is odd number in the outcome of Bucket Size then - Larger group comes first.

```
NTILE(2) OVER (ORDER BY Sales DESC)
```

Sales	NTILE
100	1
80	1
80	1
50	2
30	2

$$\text{Bucket Size} = \frac{\text{Number of Rows}}{\text{Number of Buckets}}$$

$$2 = \frac{5}{2}$$

SQL RULE

Larger groups come first